Lab 6: RLC resonance, Bandwidth and Quality Factor

Objectives

Investigate the resonance phenomena in an RLC circuit. Analyze the Resonant Frequency and Bandwidth of the given circuit and determine the effect of the load resistance on the Quality Factor.

Resonance in a series RLC circuit

Resonance is a condition in an RLC circuit in which the capacitive and inductive reactances are equal in magnitude, thereby resulting in a purely resistive impedance. Consider the series RLC circuit in Figure 1.

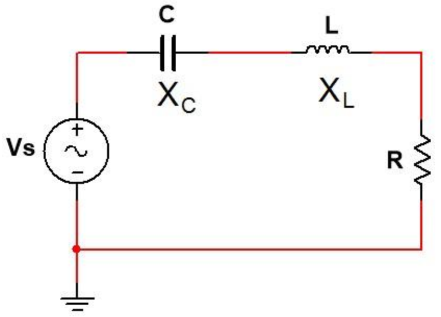

Fig.1 Series RLC circuit

Here, both X_C and X_L are frequency dependent. However, X_C is inversely proportional to frequency and X_L is directly proportional. That is, as the frequency is increased, X_C decreases and X_L increases. This means there must be a source frequency for which $X_C = X_L$. A circuit operating in that frequency is said to be in resonance condition.At the resonance condition, $X_C = X_L$, which means $\frac{1}{j\omega C} = j\omega L$. This expression can be simplified to obtain the following expression for the resonance frequency, f_0.

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

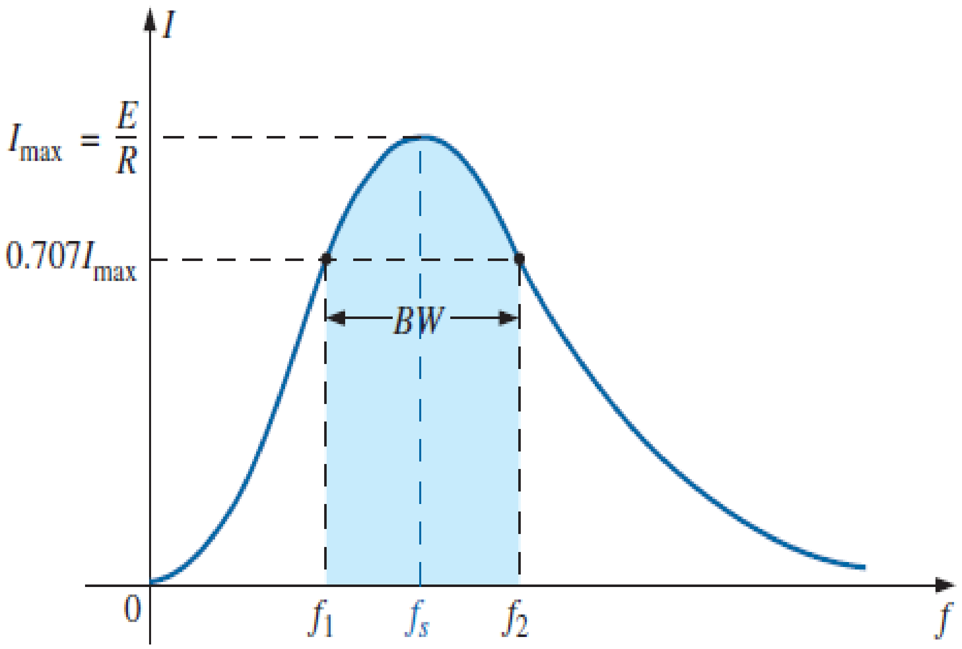

Fig.2

Now, the impedance in the above circuit can be expressed as: $Z = R + jX_L - jX_C$. Therefore, at resonance condition, the impedance of the circuit is purely resistive, which means that the entire source voltage is across R. At any other operating frequency, the voltage across R will be lower due to the non-zero reactance contributing to the total impedance.

Bandwidth and Quality Factor

If the series RLC circuit is driven by a variable frequency at a constant voltage, then the magnitude of the voltage, V is proportional to the impedance, Z. Therefore, at resonance the power absorbed by the circuit must be at its maximum value.

If we now reduce or increase the frequency until the average power absorbed by the resistor in the series resonance circuit is half that of its maximum value at resonance, we produce two frequency points called the half-power points or half-power frequencies. These points give us a voltage value that is 70.7% of its maximum resonant value. That is, at the upper and lower cut-off frequencies, say f_1 and f_2, $V = 0.707V_{max}$. The difference between these two points ($f_1 - f_2$) is called the Bandwidth (BW) of the circuit. This bandwidth is also known as the half-power bandwidth.

The "sharpness" of the resonance in a resonant circuit is measured quantitatively by the Quality Factor (Q).The quality factor relates the maximum or peak energy stored in the circuit (the reactance) to the energy dissipated (the resistance) during each cycle of oscillation meaning that it is a ratio of resonant frequency to bandwidth $(Q = \frac{f_0}{BW})$ and the higher the circuit Q, the smaller the bandwidth.

Apparatus

Components	Instruments
• Resistors: 1×100Ω (R1), 1×200Ω (R2) • Capacitors: 1×0.1µF • Inductor: 1x330µH	• 1× Trainer Board • 1× Function Generator • 1× Dual Channel Oscilloscope • Connecting wires and probes

Constructing the circuit :

1. Construct the circuit shown in **Fig.3** on the bread board. Use minimal wires.
2. Set 1 kHz Frequency and 3V peak Amplitude (6V peak to peak) in the Function Generator.
3. Connect Channel 1 of the oscilloscope across the source V_S (positive red port to node „a" and negative black port to node „0" i.e. ground).
4. Observe the generated signal on the oscilloscope screen and fine tune the frequency and amplitude of the input signal generated from the function generator to match the nominal values. Always set the amplitude after setting the frequency because changing the frequency of a non-ideal source might alter the amplitude.

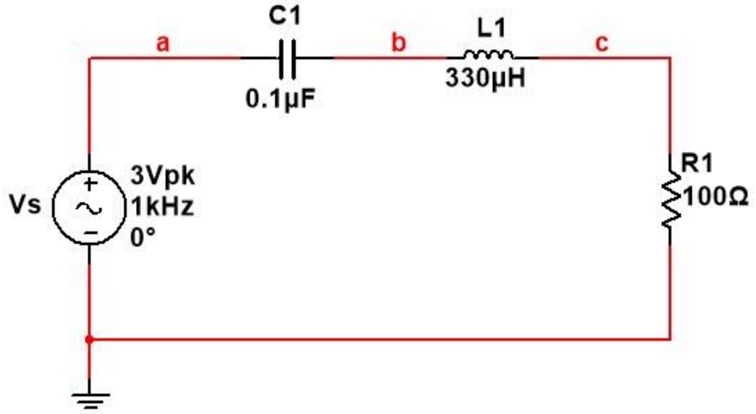

Fig.3 Series RLC circuit

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

= 1/[6.28*$\sqrt{(330X10^{-6}X0.1X10^{-6})}$]
=(1X10^{6})/36.07 Hz
=27.7 KHz

$$\boxed{f_1 = \frac{1}{2\pi}\left[-\frac{R}{2L} + \frac{1}{2}\sqrt{\left(\frac{R}{L}\right)^2 + \frac{4}{LC}}\right]} \qquad \text{(Hz)}$$

= (1/6.28)[(-100/660X10^{-6})+0.5$\sqrt{(100/330X10^{-6})^2+4/(330X10^{-6})(0.1X10^{-6})}$]

= 0.159[-0.1515X10^{6}+0.5$\sqrt{0.091X(10^6)^2+0.1212X(10^6)^2}$]

= 0.159[-0.1515X10^{6}+0.23X10^{6}]
= 0.159[0.0788X10^{6}]
= 12.53 KHz

$$\boxed{f_2 = \frac{1}{2\pi}\left[\frac{R}{2L} + \frac{1}{2}\sqrt{\left(\frac{R}{L}\right)^2 + \frac{4}{LC}}\right]} \qquad \text{(Hz)}$$

= (1/6.28)[(100/660X10^{-6})+0.5$\sqrt{(100/330X10^{-6})^2+4/(330X10^{-6})(0.1X10^{-6})}$]

= 0.159[0.1515X10^{6}+0.5$\sqrt{0.091X(10^6)^2+0.1212\ X(10^6)^2}$]

= 0.159[0.1515X10^{6}+0.23X10^{6}]
= 0.159[0.3815 X10^{6}]
= 60.65 KHz

Date:	Points:
Remarks:	

Student Information

Section:	Group:	Status:

Table 1.1: Component Values

	R_1(Ω)	**R_2** (Ω)	**C_1**(F)	**L_1**(H)
Nominal				
Measured				

Table 1.2: The magnitudes of V_R at different frequencies for a high Q circuit

Frequency (f) kHz	Peak Voltage, **V_R**(V)	I_{peak}(mA)	
1			
10			
20			
25			
27.7			
30			
40			
50			
70			
100			
120			

Table 1.3: Resonant frequency, Quality Factor and Bandwidth for a high Q circuit

	Theoretical, (a)	Experimental, (b)	Deviation (%), $\frac{a-a}{b} \times 100$
f_0 (KHz)			
f_1 (KHz)			
f_2 (KHz)			
Bandwidth ($f_2 - f_1$) **(KHz)**			
Q-factor(fo/Bandwidth)			

Table 1.4: The magnitudes of V_R at different frequencies for a low Q circuit

Frequency (f) kHz	Peak Voltage, **V_R**(V)	I_{peak}(mA)	
1			
20			
40			
60			
80			
100			
120			
140			
160			
180			
200			

Table 1.5: Resonant frequency, Quality Factor and Bandwidth for a low Q circuit

	Theoretical, (a)	Experimental, (b)	Deviation (%), $\frac{a-b}{a} \times 100$
f_0			
f_1			
f_2			
Bandwidth ($f_2 - f_1$)			
Q-factor$\left(\frac{f_0}{\text{Bandwidth}}\right)$			

Questions

1. Explain why the load voltage in an RLC circuit is maximum at resonance condition.

2. If a 5mH inductor was used instead of the 330µH one, what capacitance value would be required to keep the resonant frequency (f_0) the same as the value you obtained from your experiment?

3. How would the resonant frequency change if the 100Ω resistor was replaced with a 50Ω one? Explain your answer.

4. Use your experimental results and the graphs you obtained from the simulations to explain the concept of high and low Quality Factor in series RLC circuits.

5. Do the practical values of resonant frequency, bandwidth and quality factor you obtained confirm the theoretical values?